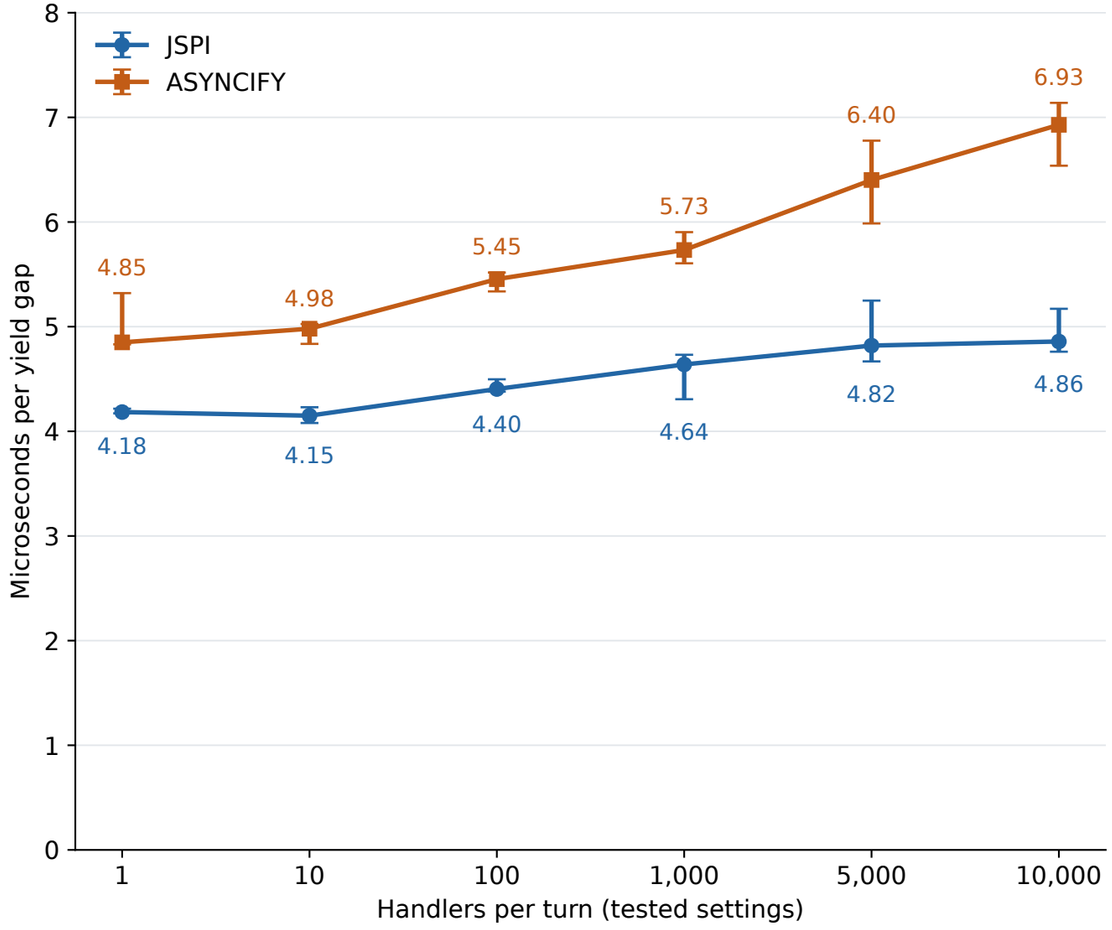
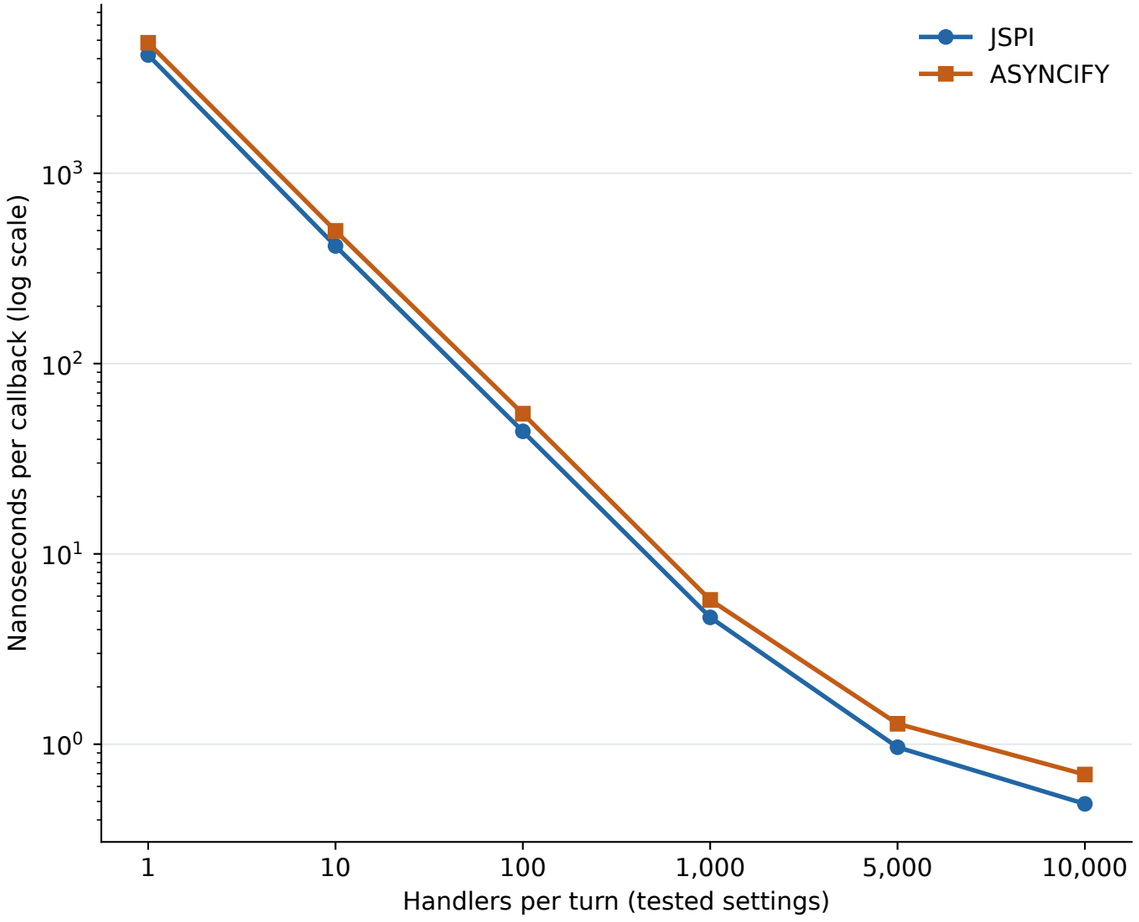
Yield latency and turn limits

Release -O3 | Exceptions enabled | Isolated page | Seven repetitions per setting

Measured boundary gap: excludes callback batch work



Derived scheduling cost per callback

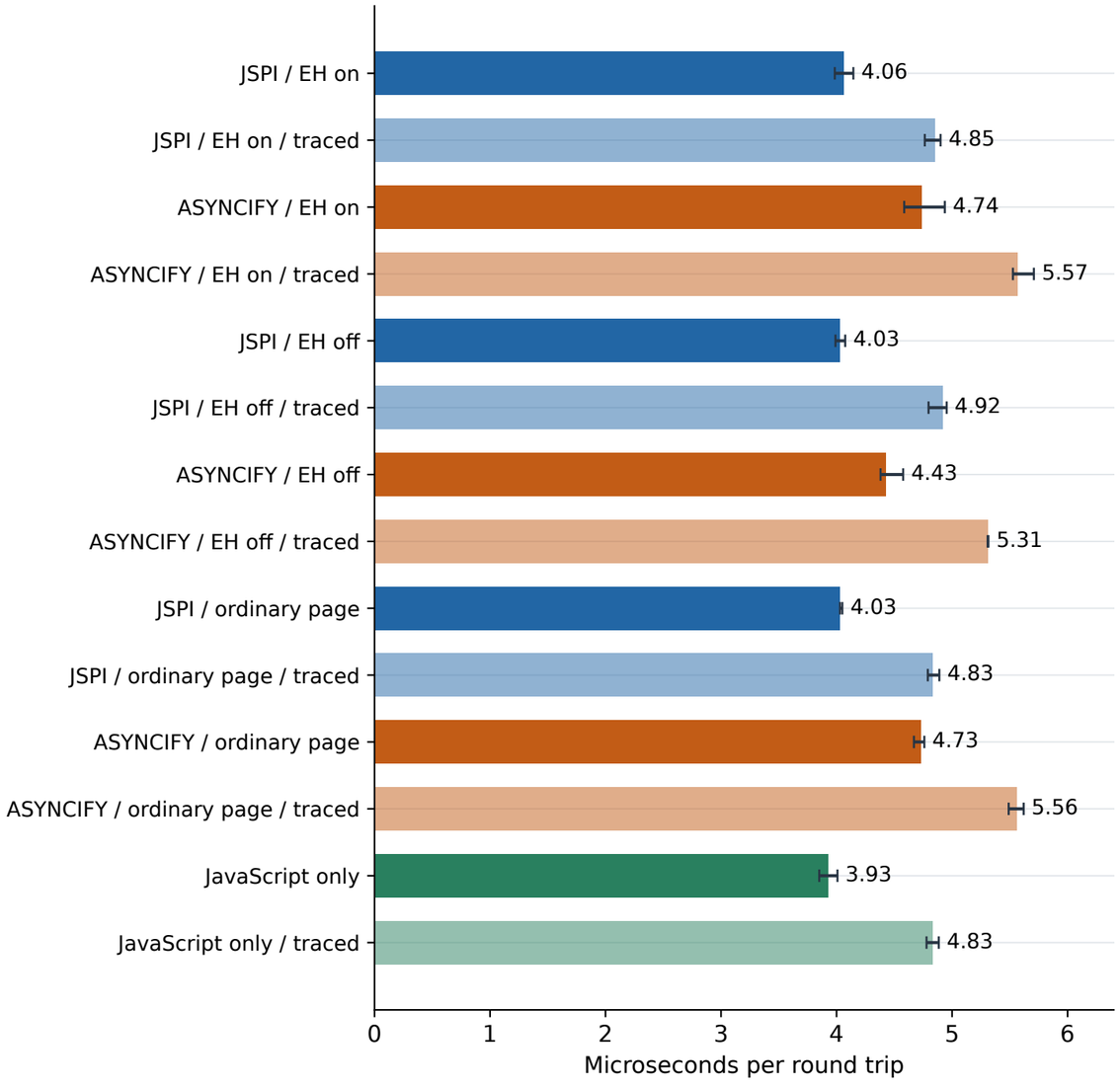


Dots: median of per-run means. Error bars: range of run means, not confidence intervals.
Amortization = measured gap / turn limit. Gaps include small dispatcher return/reentry costs.

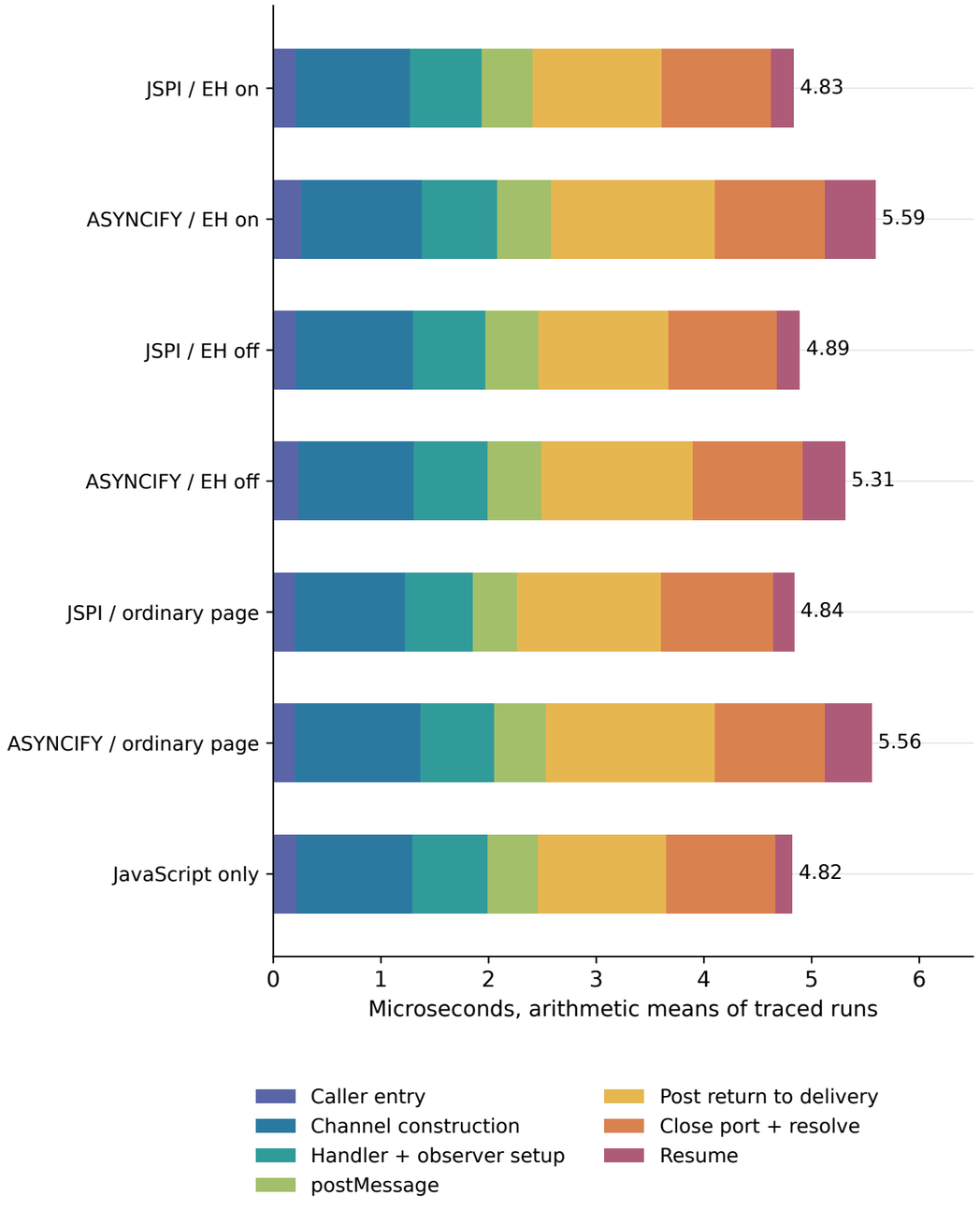
Direct round trips and observed phases

Same Promise / MessageChannel mechanism | Untraced and traced measurements kept separate

Direct latency: median of run means



Instrumented phases: means add to the total

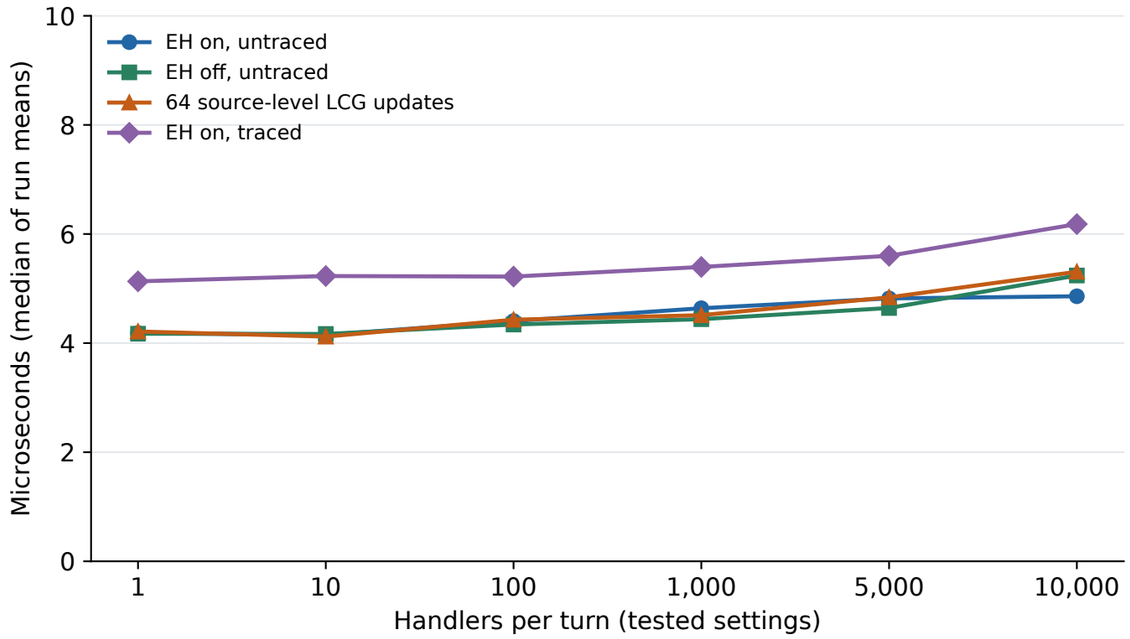


Primary and JS-only controls: 7 runs; EH-off and ordinary-page controls: 3 runs.
The observer adds cost. Post-to-delivery includes suspension and scheduling, not pure queue wait.
Sub-resolution phase values are aggregate estimates; individual clock steps are 5 or 100 us.

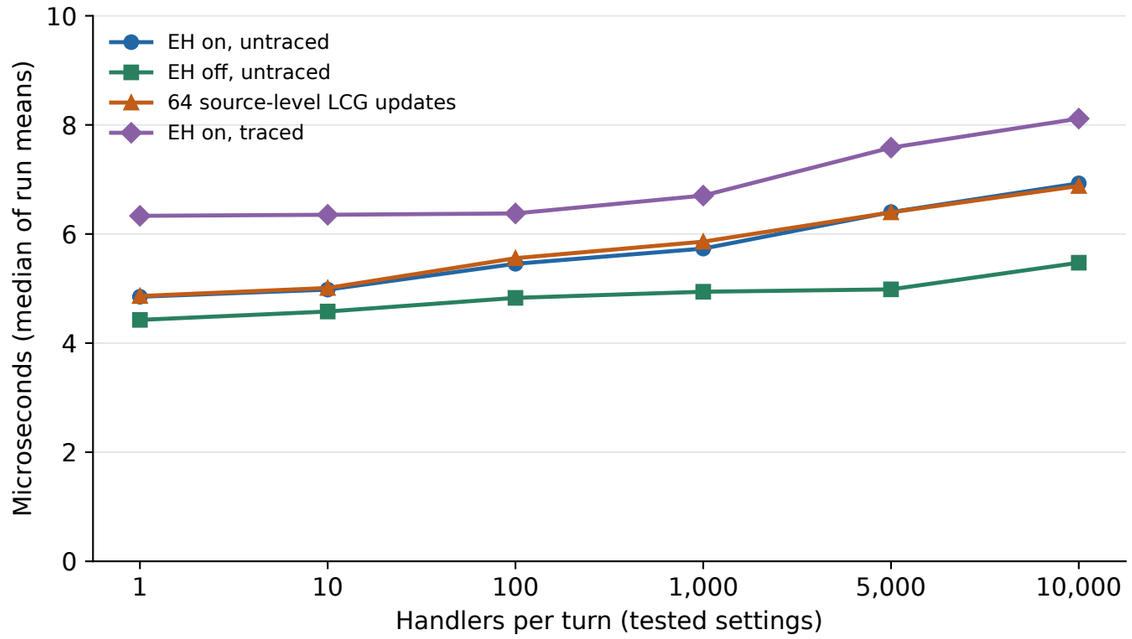
Chain controls and whole-program comparisons

All six turn limits | Same callback and timestamp counts within each control pair

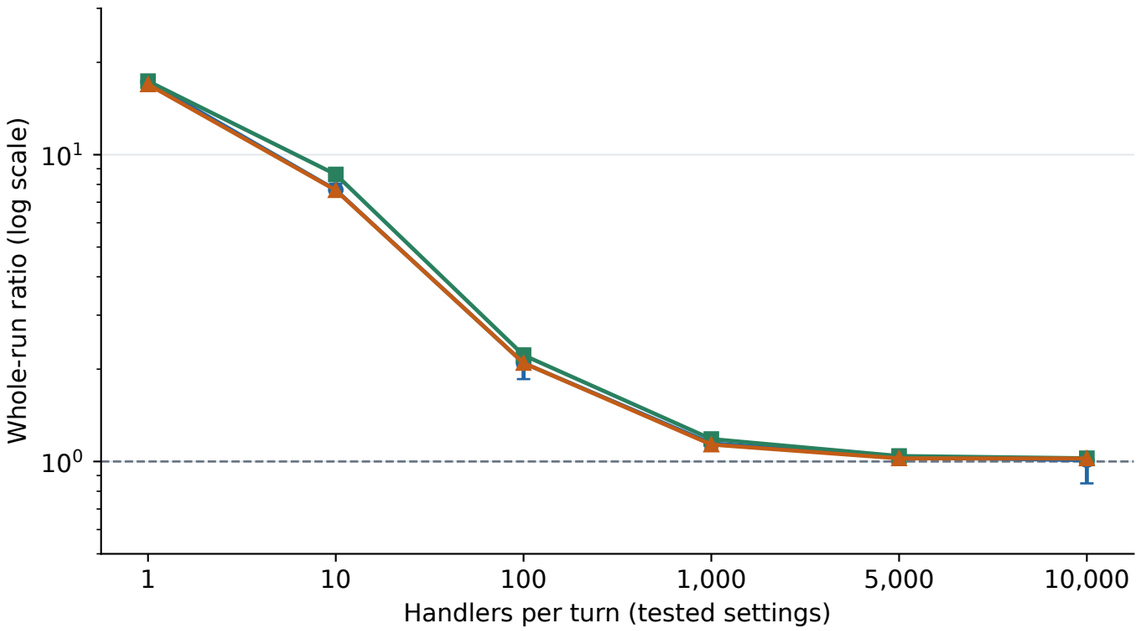
JSPI: measured yield boundary gaps



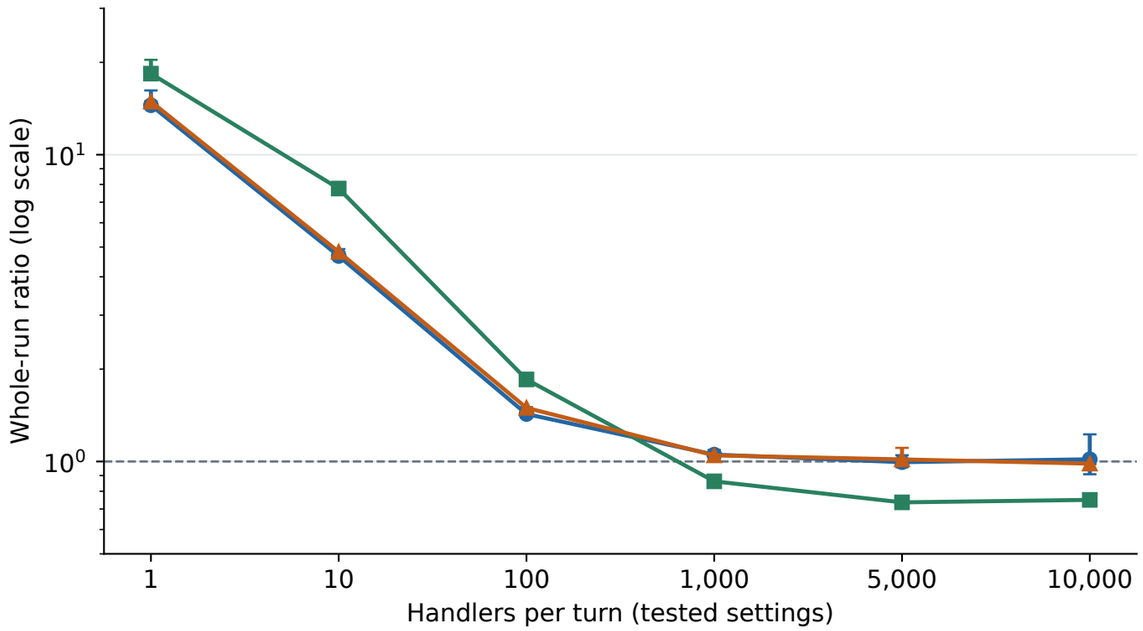
ASYNCIFY: measured yield boundary gaps



JSPI: elapsed-time ratio vs no periodic yields



ASYNCIFY: elapsed-time ratio vs no periodic yields

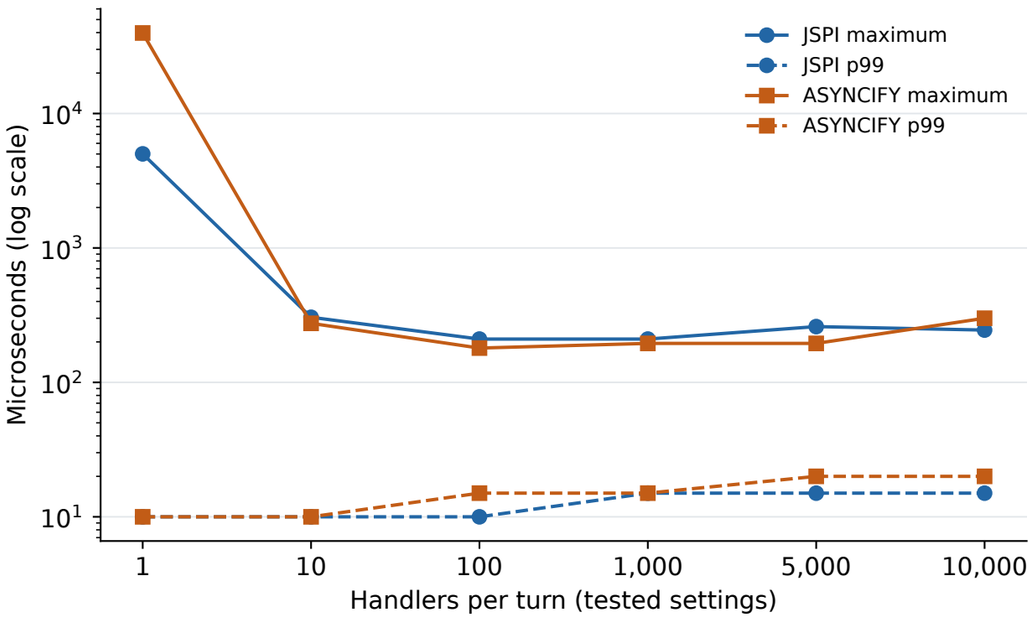


Ratios below 1 are observed whole-program outcomes, not negative yield costs. Runtime effects confound subtraction.
Primary: 7 runs; other groups: 3. Error bars show the range of paired ratios.
LCG updates are source-level work, not a guarantee of 64 instructions after optimization.

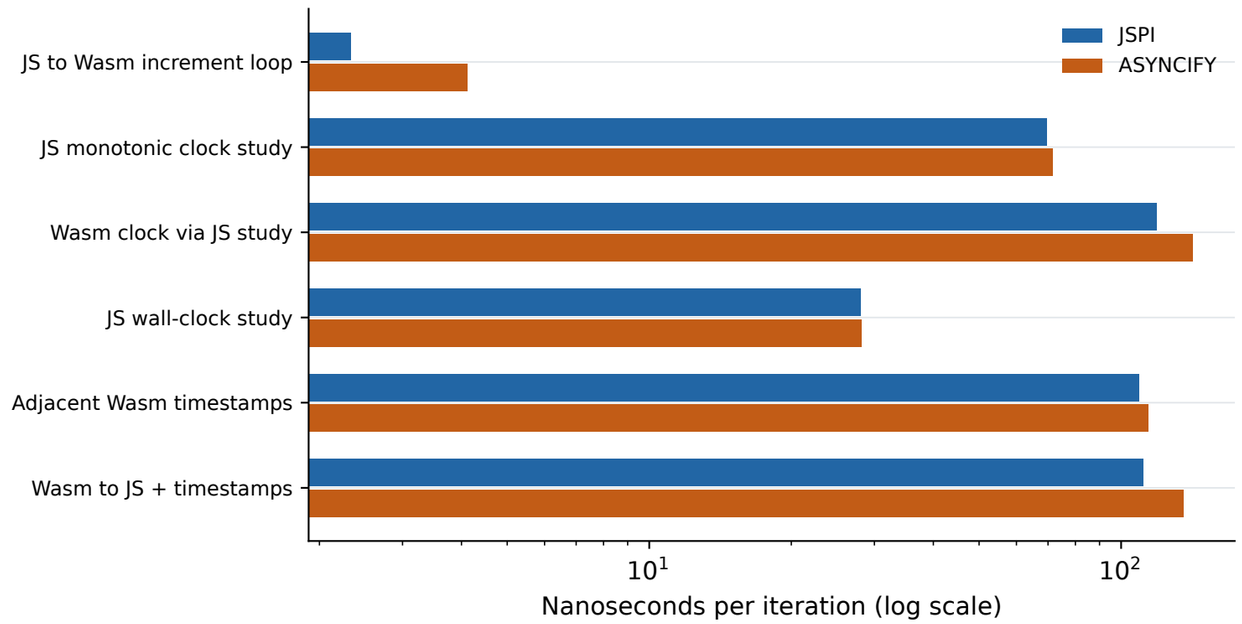
Tails, clocks and measurement resolution

Extreme observations are retained | A quantized zero is not proof of a zero-cost operation

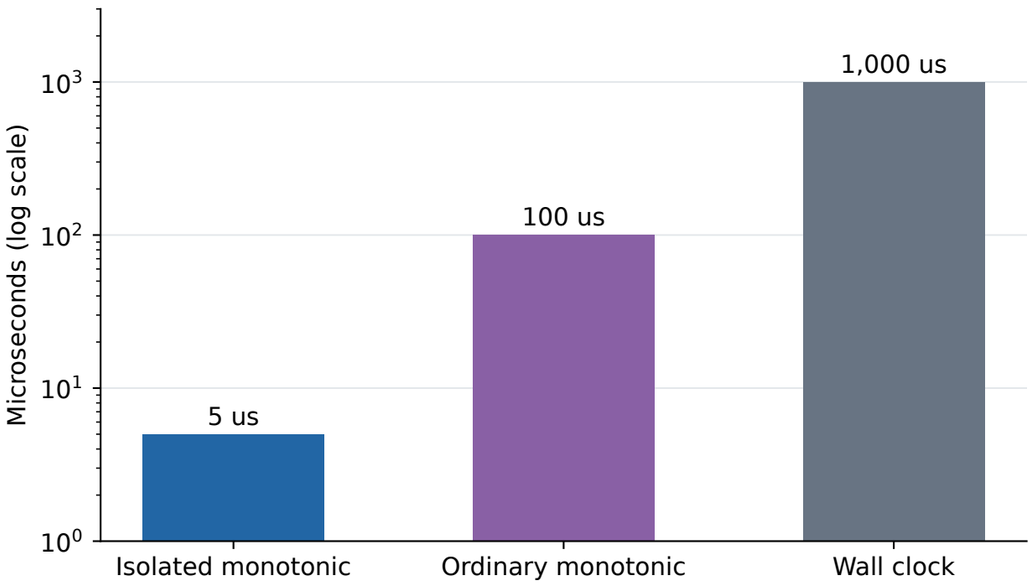
Chain tails: p99 and largest observed gap



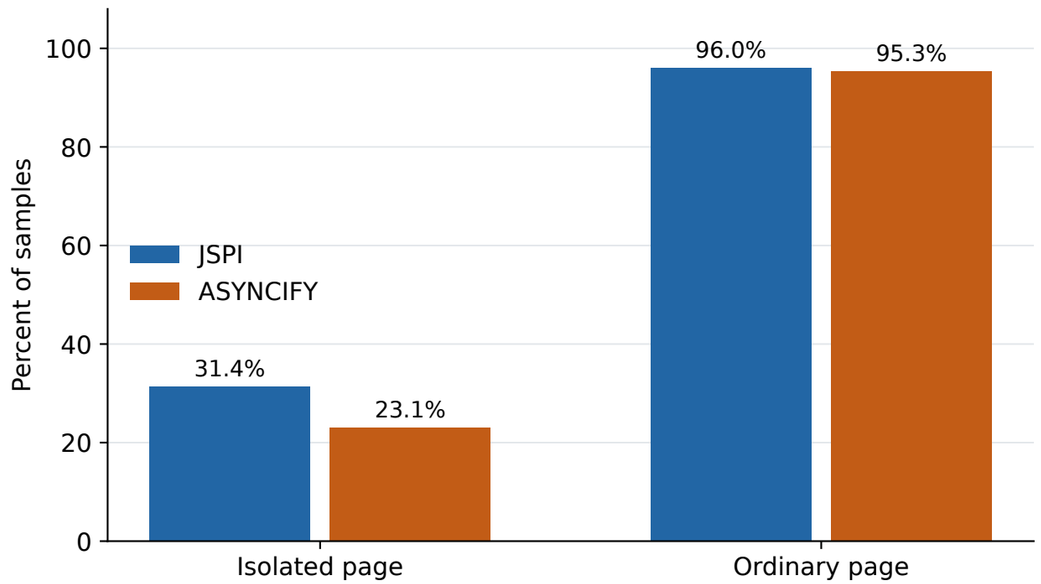
Synchronous and clock-read controls



Observed positive clock steps



Direct samples recorded as zero



Clock studies include loop overhead; the trivial synchronous export may be inlined. Wall-clock step: where observed.
Wasm and JS monotonic clocks share performance.now(). Maximums are observations, not guaranteed latency bounds.